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Market Design

Matching Markets:

- markets in which prices are not doing all work
↳ i.e. college admissions, labor mkt, marriage
↳ have to be chosen
- rely on other institutions
- rules:
 - ↳ design to have "good outcomes"
- need to understand participant responses/reactions

Kurt Sweat - Waitlist Matching (Organ donation)

Two-Sided Matching:

- set of firms: $\{f_1, \dots, f_n\} = F$
- set of workers: $\{w_1, \dots, w_n\} = W$
- Each w_i has a strict preference $\succ_{w_i}$ over $F \cup \{\emptyset\}$
- Each f_i has a strict preference $\succ_{f_i}$ over $W \cup \{\emptyset\}$
- a matching $\mu: W \cup F \rightarrow W \cup F \cup \{\emptyset\}$
 - ↳ is such that $|\mu(w)| = 1, |\mu(f)| = 1$
 $\mu(w) \in F \cup \{\emptyset\}, \mu(f) \in W \cup \{\emptyset\}$
 $\mu(\mu(f)) = f$
- ↳ match to one thing
match to other or nothing
matching is consistent
- use Roth residency match literature in hiring tournament
 - ↳ unraveling

- Given a matching μ , A blocking pair (w, f) if $w \succ_f \mu(f)$ and $f \succ_w \mu(w)$
- a matching is stable if there are no blocking pairs
- report truthfully bc Deferred Acceptance
 - Gale & Shapley (62) - Stable Marriage Model